

AGRICULTURAL SUPPLY CHAIN & POST-HARVEST LOSS MITIGATION

TEAM MEMBERS

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PROBLEM STATEMENT

In Sub-Saharan AgTech and logistics analytics, the problem of high post-harvest perishability across smallholder horticultural supply chains leads to severe crop spoilage (30–50% produce loss), depressed farmgate earnings, and artificial market price volatility.

Currently, static distribution scheduling and fixed delivery routes are used, which fail because they operate without real-time integration of ambient thermal variables, corridor transport delays, and dynamic urban market price fluctuations.

We propose using geospatial network optimization, predictive regression modeling, and spatial-temporal feature engineering to predict shelf-life decay and dynamically route high-value perishable produce to optimal wholesale hubs, aiming to achieve a 15–25% reduction in total transit duration and a measurable reduction in overall post-harvest revenue loss.

DOMAIN AND CONTEXT

Agricultural Supply Chain, Logistics Analytics, and GIS within the Sub-Saharan AgTech sector (focusing on smallholder horticultural supply chains and farmgate-to-wholesale transport corridors).

DATASET REQUIREMENTS & SCHEMA OVERVIEW

The analysis will leverage a combination of open-source agricultural indicators, meteorological datasets, and open logistics telemetry:

- **FAOSTAT / Regional Ag-Data:** Historical post-harvest loss indices and regional production volumes.
- **Climate Data (CHIRPS / ERA5):** Ambient daily temperature, humidity, and rainfall indices along transport corridors.
- **Logistics Telemetry (Simulated/Open Source):** Transit duration, vehicle speed, stopover times, and transport vehicle types.

SUCCESS METRICS

- **Prediction:** We should be able to predict the shelf-life decay percentage with high accuracy. We plan on evaluating using standard regression metrics (RMSE, R^2 , and MAE).
- **Logistics:** Assist with reducing the total transit duration for perishable shipments by 15% - 25%.
- **Spoilage reduction:** measurable reduction in overall post-harvest crop spoilage and revenue loss for smallholder farmer cooperatives.

INITIAL TIMELINE

- **Weeks 7–8 (Data Ingestion & Feature Engineering):** Ingest and clean FAOSTAT, ERA5/CHIRPS, and synthetic telemetry datasets using Pandas and GeoPandas; construct spatial-temporal features, including thermal heat exposure indicators and route friction indices via NetworkX and Haversine metrics.
- **Weeks 9–10 (Model Training & Geospatial Optimization):** Train and evaluate regression models (Random Forest, XGBoost) and time-series models (Prophet) for spoilage decay prediction; develop dynamic shortest-path and yield-maximization routing algorithms using NetworkX and Linear Programming (PuLP).
- **Weeks 11–12 (Dashboard Deployment & Capstone Delivery):** Build an interactive analytics dashboard (Streamlit / Plotly Dash / Flutter) visualizing loss hotspots, real-time market price margins, and dynamic dispatch routes; finalize codebase documentation and submit the analytical report.